

Python BDD Restful Automation

Assignment 3: Selenium Page Object Model in the Python Behave Framework

1. Problem Statement: -

Leverage Behave framework for test data driven approach combined with Page Object Model (POM).

2. Objective: -

Debug and optimize a web application automation using behave framework and Gherkin language by incorporating POM architecture.

3. Tools, Software, and Concepts Used: -

Behave, Python, Selenium WebDriver.

4. Source Code / Implementation: -

pages/login_page.py:-

```
from selenium.webdriver.common.by import By
class LoginPage:
    def __init__(self, driver):
        self.driver = driver
        self.user_input = (By.ID, 'user-name')
        self.pwd_input = (By.ID, 'password')
        self.login_btn = (By.ID, 'login-button')
    def login(self, username, password):
        self.driver.find_element(*self.user_input).send_keys(username)
        self.driver.find_element(*self.pwd_input).send_keys(password)
        self.driver.find_element(*self.login_btn).click()
```

features/pom_login.feature:-

Feature: POM Integration Testing

Scenario: Login using Page Object Model

Given I load the application

When I authenticate with POM

Then I am logged in securely

steps/pom_steps.py:-

```
from behave import given, when, then
from selenium import webdriver
```

```

from pages.login_page import LoginPage
@given('I load the application')
def step_impl(context):
    context.driver = webdriver.Chrome()
    context.driver.get('https://www.saucedemo.com/')
    context.login_page = LoginPage(context.driver)
@when('I authenticate with POM')
def step_impl(context):
    context.login_page.login('standard_user', 'secret_sauce')
@then('I am logged in securely')
def step_impl(context):
    assert 'inventory' in context.driver.current_url
    context.driver.quit()

```

5. Output & Screenshots: -

```

PS D:\Wipro Selenium> behave features/pom_login.feature
USING RUNNER: behave.runner:Runner
Feature: POM Integration Testing # features/pom_login.feature:1

  Scenario: Login using Page Object Model # features/pom_login.feature:3
    Given I load the application # features/steps/pom_steps.py:5 2.172s
    When I authenticate with POM # features/steps/pom_steps.py:11 0.177s
    Then I am logged in securely # features/steps/pom_steps.py:15 2.182s

1 feature passed, 0 failed, 0 skipped
1 scenario passed, 0 failed, 0 skipped
3 steps passed, 0 failed, 0 skipped
Took 0min 4.530s

```

Fig.1: Terminal output showing the successful execution of a Behave BDD login test using the Page Object Model.

6. Result, Observation, and Conclusion: -

- Result: Step definitions were successfully decoupled from the raw WebDriver locator logic using a dedicated Page class.
- Observation: If the UI changes, only the LoginPage class needs modification; the Behave step files remain unchanged.
- Conclusion: Combining POM with Behave creates a highly maintainable, scalable, and easy-to-read automation repository.